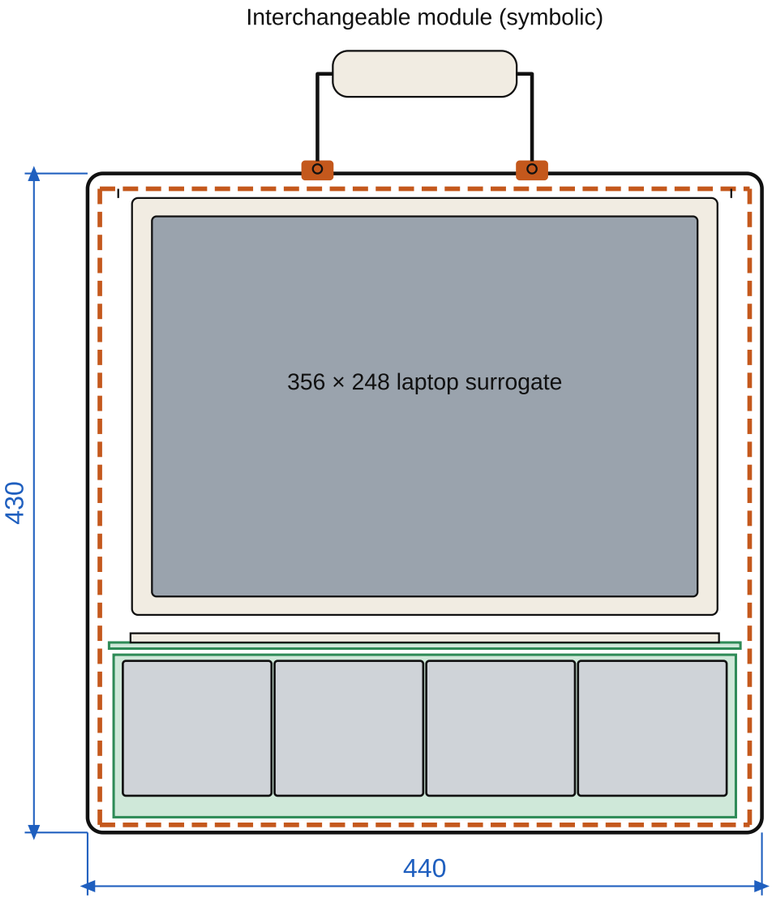
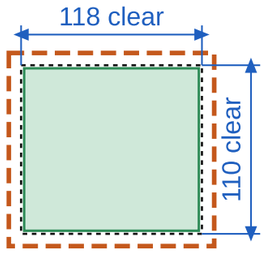
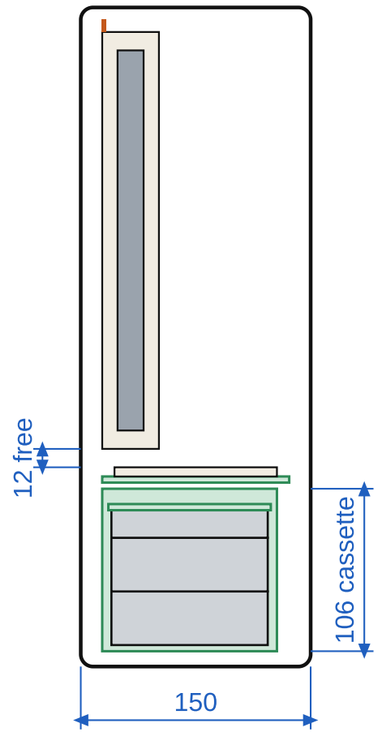


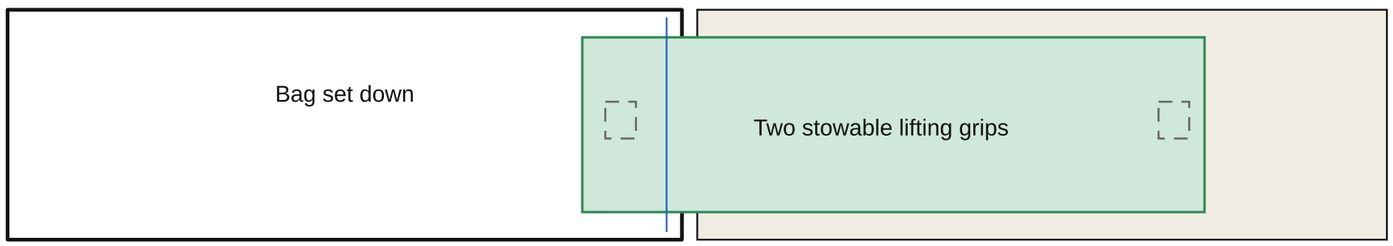

FRONT — 1:4 AT A3

END HATCH — 1:4




SECTION — 1:4




PROPOSAL A: SUPPORTED SLIDE — NOT SELECTED

Travel stop before loss of runner support; deliberate final release only after support.

## LAPTOP BAG — HEAVY BRIEF

For office workers who lift heavy. Start light; 30 kg total maximum, not a required goal. Targets untested.

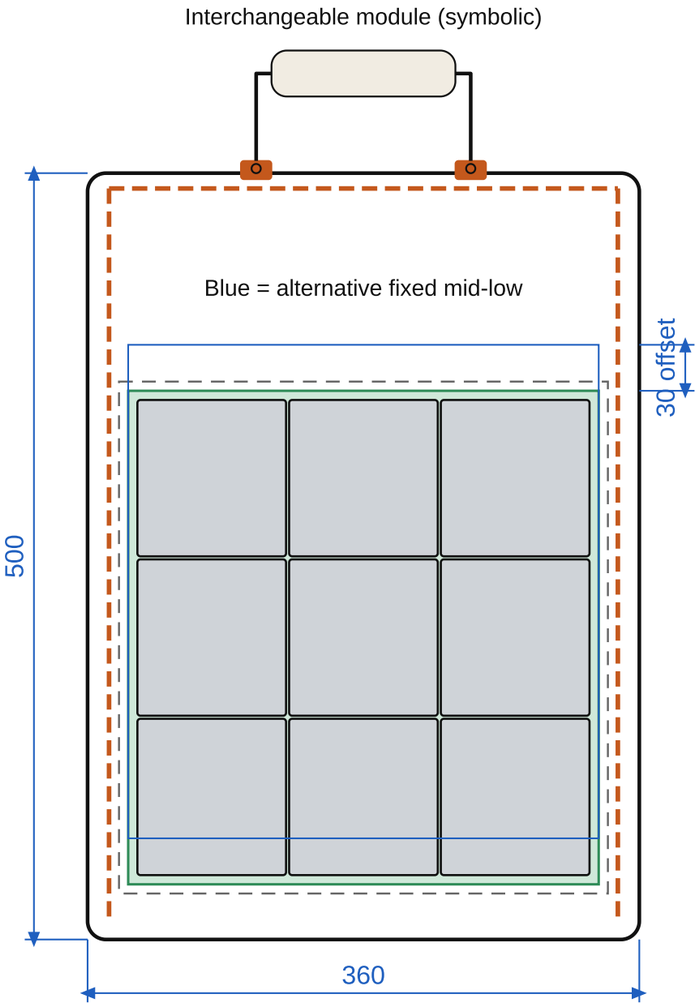
1. 440 W × 430 H × 150 D body. 406 L × 114 D × 106 H cassette; clear bay 414 × 122 × 110.
2. Clear hatch 118 D × 110 H after zip and lining intrusion. End-frame load path must route around hatch, not cross extraction path.
3. Base stack: 10 cradle + 106 cassette + 4 clearance + 4 barrier + 6 impact pad + 12 free gap + 272 padded sleeve + 16 top = 430.
4. Suspended sleeve clear 370 × 260 × 25; outer 382 × 272 × 37. Replaceable pads. Test hard bottom-out with a surrogate, not a real laptop.
5. Frame-mounted modular fabric or standard rigid handle. Captive bolts POC. Double weight handle meaning/geometry TBD; see sheet 3.
6. Supported single cassette vs two smaller cassettes stays OPEN. Split loses continuous long-bar fit. Neither is a final mechanism decision.
7. 88 mm stack + 4 clamp + 6 reserve. Two full + one half = 6.25 kg/module; four = 25 kg, over example 24 kg allowance.
8. 8 full + 3 half = 23.75 kg nominal example. Actual complete mass, including tray and handle, must be ≤30 kg. No automatic volume limiter.

**PROVISIONAL PACKAGING ONLY — NOT ENGINEERING VALIDATED**

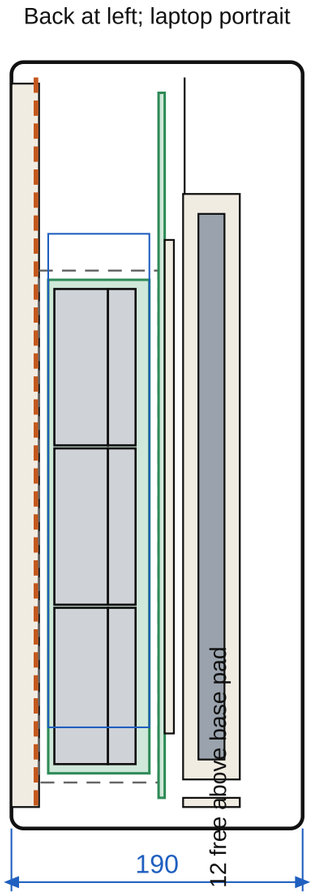
Blue: dimensions · orange: frame/receivers · green: barrier/enclosure · grey: weight

SVG / PNG / PDF exports regenerated from current generator. See tech-pack v0.2.

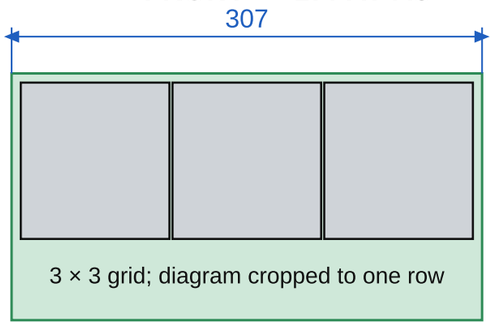
| HEAVY BAG — LAPTOP BAG | |
|---|---|
| Product-design / POC revision v0.2<br>Concept sketch · dims in mm · confirm with sample | Sheet 1 of 3<br>Views as labelled |




FRONT — 1:4 AT A3




SECTION — 1:4




### Carrier overall 307 W × 322 H × 66 D

Clear cavity 299 × 314 × 58; 53 stack +4 clamp +1 reserve.

Bay 319 × 334 × 78. Side structure/textile 20.5 each.

LOW bottom36 / top358; MID-LOW bottom66 / top388.

Separate fixed layouts, no adjustable slider in POC.

9 × 2.5 =22.5 kg; +1 ×1.25 =23.75 kg, not24.

Heavy top-lift: grounded bag, two grips, rig-test first.

## BACKPACK — BIG PACK

30 kg complete mass ceiling; start lighter. Hand carry first; worn mode is secondary, not primary rucking.

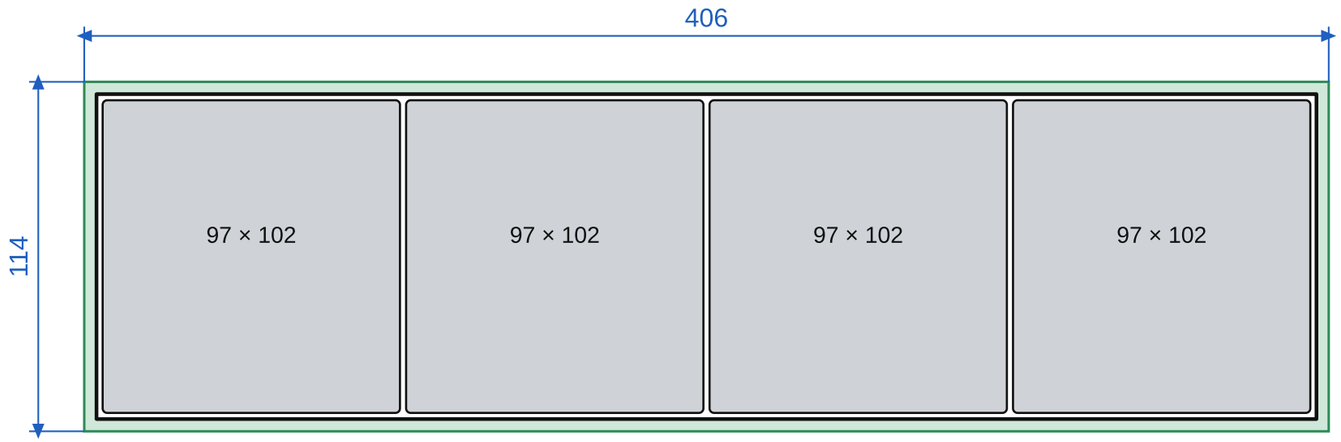
1. 360 W × 500 H × 190 D body. Carrier 307 × 322 × 66. Old 310-wide bag / 300-high zone did not allow finished parts and frame.
2. Low vs fixed mid-low is a PROPOSAL comparison, not a selected position. Compare hand tilt and shoulder pressure at equal total mass.
3. Back-to-front: 18 pad/frame + 78 bay + 4 barrier + 12 gap + 37 sleeve + 41 main/front =190. Gap includes 6 replaceable pads, 6 free.
4. Portrait laptop sleeve clear 260 W × 370 H × 25 D; padded outer 272 × 382 × 37. Bottom32: 14 base +6 pad +12 free gap.
5. Suspend sleeve; test lateral and base impact with surrogate. No load transfer from carrier or frame to laptop.
6. Stow harness for hand carry. Top-loading carrier path, zipped opening and retention need real-part insertion tests. Two lifting grips stow inside carrier envelope.
7. Modular handle receivers and captive-bolt POC swaps. Double variant TBD. Handle location must be checked against actual loaded CoM, not guessed.
8. Grid insert removable for compatible flat plates. Finished stack ≤53 incl. coating/separators; check actual SKU. Not every weight source reaches24 kg.

**PROVISIONAL PACKAGING ONLY — NOT ENGINEERING VALIDATED**

Blue: dimensions · orange: frame/receivers · green: barrier/enclosure · grey: weight

SVG / PNG / PDF exports regenerated from current generator. See tech-pack v0.2.

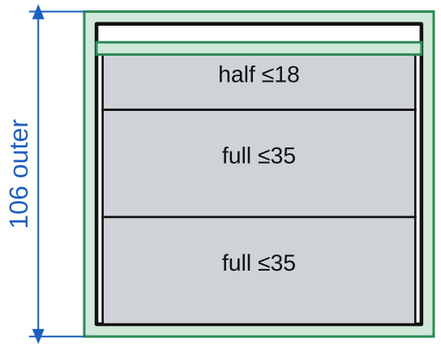
| HEAVY BAG — BIG PACK | |
|---|---|
| Product-design / POC revision v0.2<br>Concept sketch · dims in mm · confirm with sample | Sheet 2 of 3<br>Views as labelled |




## LAPTOP CASSETTE TOP — 1:2 AT A3

398 cavity =4×97 +3×2 +2×2; depth106 =102 +2×2

4 clamp +6 reserve above stack




## END SECTION — 1:2

### Mass and finished dimensions are separate

Core reference95 ×100 ×33 steel ≈2.461 kg before radii/coating.

Nominal2.5 /1.25 kg targets: measure finished; tune core to suit.

Finished full ≤97 ×102 ×35; half ≤97 ×102 ×18.

4 ×(2.5 +2.5 +1.25) =25 kg, not24.

23.75 kg example: 8 full +3 half; weigh actual whole bag.

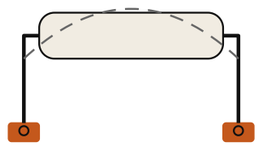
88 stack +4 clamp +6 reserve =98 inner height.

7 sleeved bars ×12 =84 fit; 8×12 =96 do NOT fit.

Lower stacks require secured spacers and clamp testing.

## MODULAR HANDLE CONCEPT — NTS

Interchangeable module (symbolic)



Shared frame receivers: provisional140 centres

Fabric module (dashed) or standard rigid module

120 grip length; ≥45 knuckle clearance; trial30/35/40 grip

Captive bolts; swap unloaded with tool; fastening details TBD

**DOUBLE WEIGHT HANDLE: MEANING / GEOMETRY TBD**

## PROVISIONAL PACKAGING ONLY — NOT ENGINEERING VALIDATED

Blue: dimensions · orange: frame/receivers · green: barrier/enclosure · grey: weight

SVG / PNG / PDF exports regenerated from current generator. See tech-pack v0.2.

## WEIGHT INTERFACE — SOURCE LIMITS

One shared interface goal, two enclosure shapes. No universal-fit or universal24 kg claim. Complete bag ≤30 kg.

1. Finished95 ×100 core coating allowance gives ≤97 ×102 footprint. 2 mm edge and intermodule gaps. Source limits in tech pack v0.2.
2. Big Pack cavity299 ×314 ×58; 3×97 +2×2 +2×2 =299; 3×102 +2×2 +2×2 =314. 53 stack +4 clamp +1 reserve.
3. Raw bar laptop380 ×100 ×10 ≈2.983 kg. Finished sleeve ≤384 ×104 ×12. Seven layers ≈20.88 kg steel; add sleeve tare.
4. Big Pack raw285 ×100 ×10 ≈2.237 kg; sleeved ≤289 ×104 ×12. 2 rows ×4 layers =8 pieces, ≈17.90 kg steel. Three rows need320 high with gaps/edges; do not fit. Secure unused space.
5. Shot theoretical upper bounds before shell losses: laptop13.7–17.8 kg, Big Pack18.4–23.9 kg at3.7–4.8 kg/L. Real capacity is lower.
6. Water/sand much lighter. Long shells must fit cavity and stack including caps; nominal outside shell volume is not fill capacity.
7. Ruck plate229 ×292 is a candidate Big Pack footprint, insert removed. Finished stack ≤53; sleeves/clamp included. Not every SKU fits.
8. Small gym plates ≤150 diameter are Big Pack candidates only; confirm layout and stack. Laptop cavity depth106 cannot take them.
9. Proposed supports, clamps, fasteners and coatings are untested. Low-stack spacers, sealed fillers and two-grip retention need POC validation.

## HEAVY BAG — WEIGHT SYSTEM

| | |
|---|---|
| Product-design / POC revision v0.2<br>Concept sketch · dims in mm · confirm with sample | Sheet 3 of 3<br>Views as labelled |